S.No. : **82** **NME 4201**

No. of Printed Pages : 08

Following Paper ID and Roll No. to be filled in your Answer Book.		
PAPER ID : 43502	**Roll No.**	

B. Tech. Examination, 2024-25

(Even Semester)

ENGINEERING MECHANICS

Time : Three Hours] *[Maximum Marks : 60*

Note :– Attempt all questions.

SECTION – A

1. Attempt all parts of the following : $8\times1=8$

(a) Differentiate between concurrent and non-concurrent forces.

(b) Explain the term 'free body diagram'. Draw the free body diagram of a ball of weight W, placed on a horizontal surface.

(c) What is the difference between a roller support and a hinged support?

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(d) Define the terms coefficient of friction and angle of friction.

(e) Explain perfect frame, deficient frame and redundant frame.

(f) Derine radius of gyration.

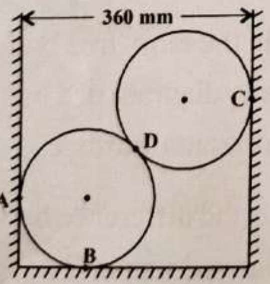
(g) State D'Alembert's principle.

(h) Define Poisson's ratio.

SECTION – B

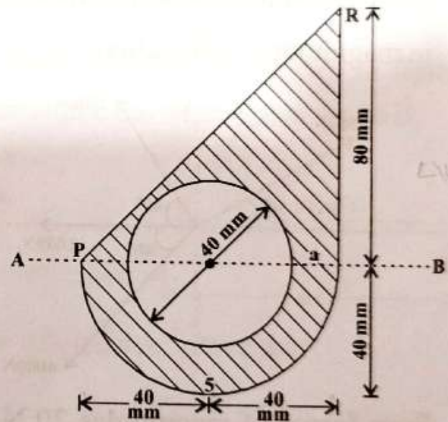
2. Attempt any two parts of the following : $2\times6=12$

(a) Two sphere, each of weight 50N and of radius 100 mm rest in a horizontal channel of width 360 mm as shown in figure below. Find the reactions on the points of contact A, B, C and D.



(b) A uniform ladder of 7m rests against a vertical wall with which it makes an angle of 45^0. The coefficient of friction between the ladder and the wall is 1/3 and that between ladder and floor is ½. If a man whose weight is one half of that of the ladder, ascends its, how high will it be when the ladder slips.

(c) Find the moment of inertia of the shaded area as shown in figure below about the axis AB.



(d) Explain the following :

(i) Normal stress

(ii) Shear stress

(iii) Strain

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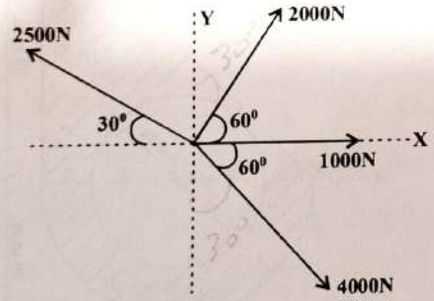
(iv) Modulus of elasticity

(v) Modulus of rigidity

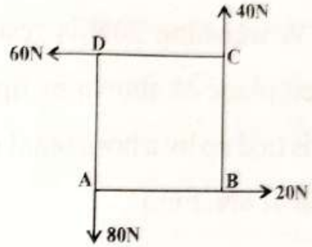
SECTION – C

Note :– Attempt all questions. Attempt any two parts from each questions. $8\times5=40$

3. (a) The four coplaner forces are acting at a point as shown in figure below. Determine the resultant in magnitude and direction.



(b) Four forces of magnitudes 20 N, 40 N, 60 N and 80 N are acting along the four sides of a square ABCD as shown in figure below. Determine the resultant moment about point A. each side of square is 2m.



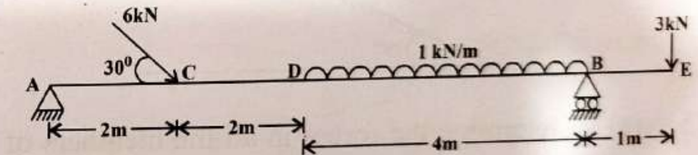
(c) Explain the following :

(i) Lami's theorem

(ii) Varignon's theorem

(iii) Parallelogram law of forces

4. (a) A beam is loaded as shown in figure below. Determine the reactions at A and B.



(b) For a belt drive, prove that :

$$\frac{T_2}{T_1} = e^{\mu\theta}.$$

where T_2 is the tension on tight side, T_1 is the tension on the slackside, μ is the coefficient of friction and θ is the angle of contact between belt and drum.

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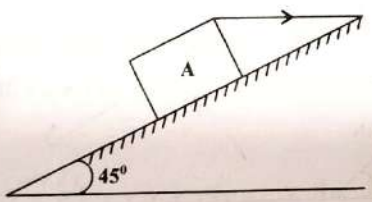
(c) Block A weishing 20N is resting on a rough inclined plane as shown in figure below. The block is tied up by a horizontal string which has tension of 6N. Find :

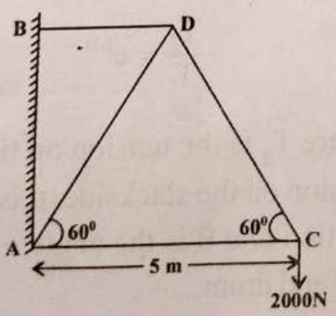
(i) The frictional force on the block.

(ii) The normal reaction of the inclined plane.

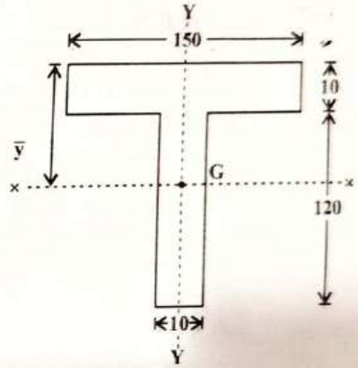
(iii) The coefficient of friction between the surface of contract.



5. (a) Determine the forces in all the members of a cantilever truss as shown in figure below.



(b) Determine the moment of inertia of the section shown in figure below about an axis passing through the centroid and parallel to the top most fibre of the section. Also determine moment of inertia about the axis of symmetry.



(c) State and prove the theorem of parallel axis.

6. (a) A force of 450 N acts on a body of mass 150 kg for 5 seconds. if the initial velocity of the body is 10 m/s. Calculate :

(i) Acceleration produced.

(ii) Distance moved by the body in 4 seconds.

(b) Draw the stress-strain diagram for mild steel and explain salient points on it.

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(c) Explain the following :

(i) Impulse momentum principles.

(ii) Work energy theorem.

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